File I/O in C++

Using Input/Output Files

A computer file

- is stored on a secondary storage device (e.g., disk);
- is permanent;
- can be used to
  - provide input data to a program
  - or receive output data from a program
  - or both;
- should reside in Project directory for easy access;
- must be opened before it is used.

Text Files

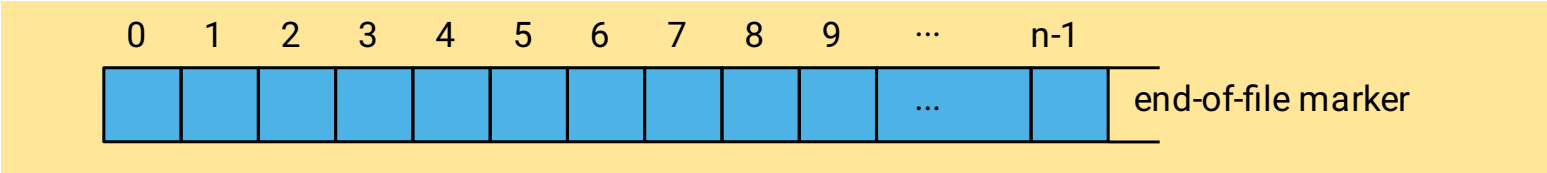
- All files are coded as long sequences of bits (0s and 1s)
- Some files are coded as sequences of ASCII character values (referred to as *text* files)
  - files are organized as bytes, with each byte being an ASCII character
- Other files are generally referred to as binary files

File Terms

- Buffer - a temporary storage area used to transfer data back and forth between memory and auxiliary storage devices
- Stream - files are manipulated with streams, a stream is a mechanism that is connected to a file that allows you to access one element at a time

C++ Files and Streams

- C++ views each files as a sequence of bytes.
- Each file ends with an ***end-of-file*** marker.



- When a file is ***opened***, an object is created and a stream is associated with the object.
- To perform file processing in C++, the header files **<iostream.h>** and **<fstream.h>** must be included.
- <fstream.> includes <ifstream> and <ofstream>

General File I/O Steps

1. Include the header file fstream in the program.
2. Declare file stream variables.
3. Associate the file stream variables with the input/output sources.
4. Open the file
5. Use the file stream variables with >>, <<, or other input/output functions.
6. Close the file.

Using Input/Output Files

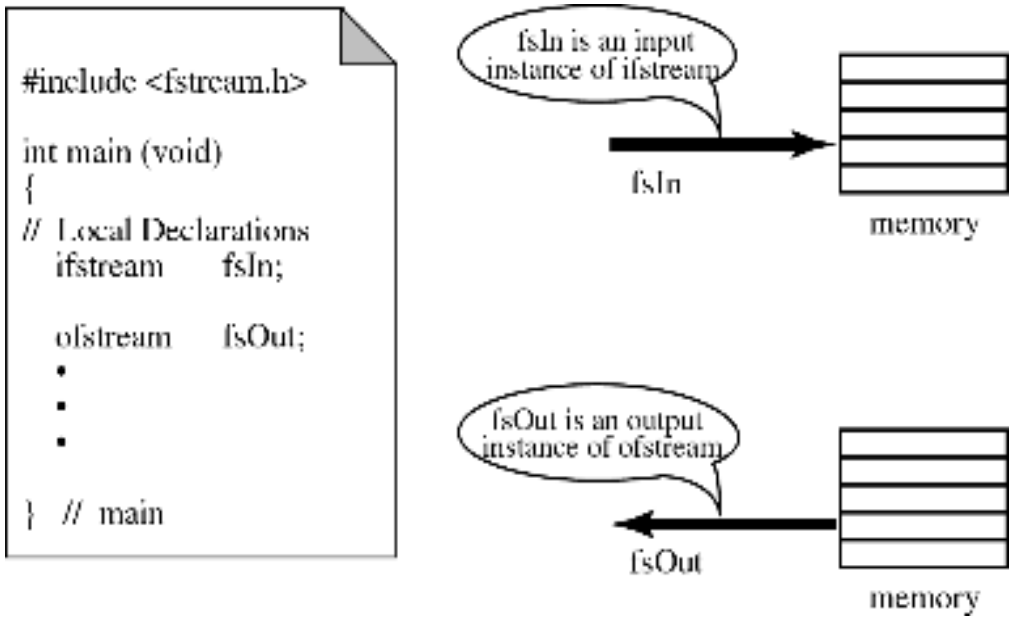
- ***stream*** - a sequence of characters
  - interactive (iostream)
    - **cin** - input stream associated with **keyboard.**
    - **cout** - output stream associated with **display**
  - file (fstream)
    - **ifstream** - defines new input stream (normally associated with a file).
    - **ofstream** - defines new output stream (normally associated with a file).

Stream I/O Library Header Files

- Note: There is no “.h” on standard header files : <fstream>
- iostream -- contains basic information required for all stream I/O operations
- fstream -- contains information for performing file I/O operations

C++ streams

```
//Add additional header files you use
#include <fstream>
int main ()
{ /* Declare file stream variables such as
 the following  */
ifstream  fsIn;//input
ofstream fsOut; // output
fstream both //input & output
//Open the files
fsIn.open("prog1.txt"); //open the input file
fsOut.open("prog2.txt"); //open the output
file
//Code for data manipulation
.
.
//Close files
fsIn.close();
fsOut.close();
return 0; }
```



Object and Member Functions

Stream handle Name

Member Function Name

input_stream.open("numbers.txt“)

Calling Object

Dot Operator

File Name
Dir:\\folder\fileName.extention
Extention (.dat, .out, .txt)

Open()

- Opening a file associates a file stream variable declared in the program with a physical file at the source, such as a disk.
- In the case of an input file:
  - the file must exist before the open statement executes.
  - If the file does not exist, the open statement fails and the input stream enters the fail state
- An output file does not have to exist before it is opened;
  - if the output file does not exist, the computer prepares an empty file for output.
  - If the designated output file already exists, by default, the old contents are erased when the file is opened.

Validate the file before trying to access

First method

By checking the stream variable;

```
If ( ! Mystream)
{
Cout << “Cannot open file.\n ”;
}
```

Second method

By using bool is_open() function.

```
If ( ! Mystream.is_open())
{
Cout << “File is not open.\n ”;
}
```

File I/O Example: Open the file with validation

First Method (use the constructor)

```
#include <fstream>
using namespace std;
int main()
{
//declare and automatically open the file
ofstream outFile("fout.txt");
// Open validation
if(! outFile) {
Cout << "Cannot open file.\n ";
return 1;
}
return 0;
}
```

Second Method (use Open function)

```
#include <fstream>
using namespace std;
int main()
{
//declare output file variable
ofstream outFile;
// open an exist file fout.txt
outFile.open("fout.txt");
// Open validation
if(! outFile.is_open() ) {
Cout << "Cannot open file.\n ";
return 1;
}
return 0;
}
```

More Input File-Related Functions

- **ifstream fsin;**
- **fsin.open(const char[] fname)**
  - connects stream **fsin** to the external file ***fname.***
- **fsin.get(char character)**
  - extracts next character from the input stream **fsin** and places it in the character variable ***character.***
- **fsin.eof()**
  - tests for the end-of-file condition.

File I/O Example: Reading

Read char by char

```
#include <iostream>
#include <fstream>

int main()
{//Declare and open a text file
ifstream openFile(“data.txt");
 char ch;
 //do until the end of file
while( ! OpenFile.eof() )
{
OpenFile.get(ch); // get one character
cout << ch;   // display the character
}
OpenFile.close(); // close the file

    return 0;
}
```

Read a line

```
#include <iostream>
#include <fstream>
#include <string>
int main()
{//Declare and open a text file
ifstream openFile("data.txt");
string line;
while(!openFile.eof())
{//fetch line from data.txt and put it in a string
getline(openFile, line);
cout << line;
}
openFile.close(); // close the file
    return 0; }
```

More Output File-Related Functions

- **ofstream fsOut;**
- **fsOut.open(const char[] fname)**
  - connects stream **fsOut** to the external file ***fname.***
- **fsOut.put(char character)**
  - inserts character ***character*** to the output stream **fsOut.**
- **fsOut.eof()**
  - tests for the end-of-file condition.

File I/O Example: Writing

First Method (use the constructor)

```
#include <fstream>
using namespace std;
int main()
{/* declare and automatically open the file*/
ofstream outFile("fout.txt");

//behave just like cout, put the word into  the file
outFile << "Hello World!";

outFile.close();

return 0;
}
```

Second Method (use Open function)

```
#include <fstream>
using namespace std;
int main()
{// declare output file variable
ofstream outFile;
// open an exist file fout.txt
   outFile.open("fout.txt”);

//behave just like cout, put the word into  the file
outFile << "Hello World!";

outFile.close();

return 0;
}
```

File Open Mode

Name	Description
ios::in	Open file to read
ios::out	Open file to write
ios::app	All the data you write, is put at the end of the file. It calls ios::out
ios::ate	All the data you write, is put at the end of the file. It does not call ios::out
ios::trunc	Deletes all previous content in the file. (empties the file)
ios::nocreate	If the file does not exists, opening it with the open() function gets impossible.
ios::noreplace	If the file exists, trying to open it with the open() function, returns an error.
ios::binary	Opens the file in binary mode.

File Open Mode

```
#include <fstream>
int main(void)
{
ofstream outFile("file1.txt", ios::out);
outFile << "That's new!\n";
outFile.close();
        Return 0;
}
```

If you want to set more than one open mode, just use the **OR** operator- |. This way:

ios::ate | ios::binary

File format

- In c++ files we (read from/ write to) them as a stream of characters

- What if I want to write or read numbers ?

Example writing to file

```
#include <iostream>
#include <fstream>
using namespace std;
void  main()
{
ofstream outFile;
// open an exist file fout.txt
    outFile.open("number.txt",ios::app);

if (!outFile.is_open())
{ cout << " problem with opening the file ";}
else
{outFile <<200 <<endl ;
cout << "done writing" <<endl;}

outFile.close();

}
```

Example Reading from file

```
#include <iostream>
#include <fstream>
#include <string>
#include <sstream>
using namespace std;
void main()
{//Declare and open a text file
ifstream INFile("number.txt");
string line;
int total=0;
while(! INFile.eof())
{
getline(INFile, line);
//converting line string to int
stringstream(line) >> total;
cout << line  <<endl;
cout <<total +1<<endl;}
INFile.close(); // close the file
 }
```